

A full-rank qubit counterexample to data processing for a trace-exponential relative functional

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

For positive-definite density matrices, consider $S(A\|B) = \log \text{Tr}[A \exp(\log A - \log B)]$. We give an explicit pair of qubit states and an amplitude-damping channel with damping parameter $1/4$ for which S strictly increases. All inputs and outputs are positive definite. A scalar spectral formula and rational interval arithmetic with proved series remainders enclose the input trace in $[7.541002855449, 7.541002855450]$ and the output trace in $[8.377802166598, 8.377802166599]$. Thus the displayed functional fails data processing already on the full-rank qubit domain. The statement concerns this precise formula; bibliographic priority is not asserted.

1 Statement and scope

Throughout, matrix logarithms are spectral logarithms, and scalar logarithms are natural. For positive-definite density matrices A, B , define

$$T(A, B) = \text{Tr}[A \exp(\log A - \log B)], \quad S(A\|B) = \log T(A, B). \quad (1)$$

The trace is positive, since it equals the trace of the positive-definite matrix $A^{1/2} \exp(\log A - \log B) A^{1/2}$. The proposed data-processing inequality is

$$S(\Phi(A)\|\Phi(B)) \leq S(A\|B) \quad (2)$$

for completely positive trace-preserving (CPTP) maps Φ , whenever the four states in this expression are positive definite.

Theorem 1. *Let*

$$\rho = \begin{pmatrix} 91/200 & (-44 + 23i)/100 \\ (-44 - 23i)/100 & 109/200 \end{pmatrix}, \quad \sigma = \begin{pmatrix} 37/40 & (-8 + 9i)/50 \\ (-8 - 9i)/50 & 3/40 \end{pmatrix}, \quad (3)$$

and set $\Phi(A) = K_0 A K_0^* + K_1 A K_1^*$, where

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{3}/2 \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & 1/2 \\ 0 & 0 \end{pmatrix}. \quad (4)$$

Then Φ is CPTP, and $\rho, \sigma, \Phi(\rho), \Phi(\sigma)$ are positive-definite density matrices. The exact enclosures are

$$\begin{aligned} 7.541002855449 &\leq T(\rho, \sigma) \leq 7.541002855450, \\ 8.377802166598 &\leq T(\Phi(\rho), \Phi(\sigma)) \leq 8.377802166599. \end{aligned} \quad (5)$$

Consequently $S(\Phi(\rho)\|\Phi(\sigma)) > S(\rho\|\sigma)$.

The conclusion refutes (2) for the formula (1). It also refutes every extension to singular states that agrees with that formula on positive-definite pairs. No value at a singular state is used.

Context and provenance. The motivating problem supplied to this study denotes (1) by S_2^Q and attributes it to Definition 3 and a data-processing discussion in a work of Meunson–Deesuwan, identified there as arXiv:2606.31205v1. That primary text was unavailable for inspection. Its precise definition, the attribution, and the reported prior open status are therefore unverified; we do not identify this calculation as a resolution of that paper’s question. The contribution established here is the explicit interior witness and its certificate for (1). No claim of novelty or priority is made. As broad adjacent methodology, the work of Zeng, Liu, and Ratnavelu [1] also concerns explicit 2×2 matrix counterexamples certified by rational inequalities. It concerns a Schatten norm inequality and is not used to justify the present data-processing claim.

2 Exact matrix and scalar reductions

The representation (4) is completely positive: after tensoring with an identity of any dimension, each summand is a congruence and preserves positive semidefiniteness. Moreover,

$$K_0^* K_0 + K_1^* K_1 = \text{diag}(1, 3/4) + \text{diag}(0, 1/4) = I,$$

so cyclicity of trace gives trace preservation. Direct multiplication yields

$$\begin{aligned}\Phi(\rho) &= \begin{pmatrix} 473/800 & \sqrt{3}(-44 + 23i)/200 \\ \sqrt{3}(-44 - 23i)/200 & 327/800 \end{pmatrix}, \\ \Phi(\sigma) &= \begin{pmatrix} 151/160 & \sqrt{3}(-8 + 9i)/100 \\ \sqrt{3}(-8 - 9i)/100 & 9/160 \end{pmatrix}.\end{aligned}$$

All four matrices are Hermitian with trace one. In the order $\rho, \sigma, \Phi(\rho), \Phi(\sigma)$, their determinants are

$$\frac{59}{40000}, \quad \frac{91}{8000}, \quad \frac{36351}{640000}, \quad \frac{1227}{128000}. \quad (6)$$

Each is positive. The two eigenvalues of each matrix have positive product and sum one, hence are both positive.

Write the Pauli matrices as

$$\tau_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \tau_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \tau_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For $x \in \mathbb{R}^3$, set $A_x = (I + x \cdot \boldsymbol{\tau})/2$. The states (3) correspond to

$$r = (-22/25, -23/50, -9/100), \quad s = (-8/25, -9/25, 17/20).$$

The channel sends (x_1, x_2, x_3) to $(\sqrt{3}x_1/2, \sqrt{3}x_2/2, 3x_3/4 + 1/4)$.

Lemma 1. *Let $0 < |x|, |y| < 1$, and write $u = |x|^2$, $v = |y|^2$, $w = x \cdot y$. Define*

$$\begin{aligned}b(u) &= \frac{1}{2\sqrt{u}} \log \frac{1 + \sqrt{u}}{1 - \sqrt{u}}, \\ q &= b(u)^2 u + b(v)^2 v - 2b(u)b(v)w, \quad c = b(u)u - b(v)v, \\ C(q) &= \sum_{n=0}^{\infty} \frac{q^n}{(2n)!}, \quad H(q) = \sum_{n=0}^{\infty} \frac{q^n}{(2n+1)!}.\end{aligned}$$

Then

$$T(A_x, A_y) = \sqrt{\frac{1-u}{1-v}} [C(q) + cH(q)]. \quad (7)$$

Proof. The eigenvalues of A_x are $(1 \pm \sqrt{u})/2$, so its spectral projectors give

$$\log A_x = \frac{1}{2} \log((1-u)/4)I + b(u)x \cdot \tau.$$

Thus $\log A_x - \log A_y = dI + t \cdot \tau$, where $d = \frac{1}{2} \log((1-u)/(1-v))$ and $t = b(u)x - b(v)y$. The Pauli relations give $(t \cdot \tau)^2 = |t|^2 I = qI$. Separating even and odd powers in the absolutely convergent exponential series therefore gives

$$\exp(dI + t \cdot \tau) = e^d [C(q)I + H(q)t \cdot \tau].$$

Now $\text{Tr } A_x = 1$ and $\text{Tr}(A_x(t \cdot \tau)) = x \cdot t = c$. Taking the trace proves (7). No commutativity of A_x and A_y is assumed. The series also gives $H(0) = 1$. $\square$

The exact arguments for the witness are

	u	v	w
input	9941/10000	1909/2000	3707/10000
output	123649/160000	30773/32000	79579/160000

(8)

In particular the hypotheses of Lemma 1 hold for both pairs.

3 Rational certification and conclusion

Here and below, every terminating decimal used as an interval endpoint denotes an exact rational number. The calculation evaluates (7) on the two triples (8) using integer endpoints on the grid $10^{-40}\mathbb{Z}$. Addition and subtraction are exact on this grid. For multiplication and division, all four endpoint combinations are considered and the extremal results are rounded outward by integer floor and ceiling; every divisor interval is checked to exclude zero. For $[L/D, U/D]$ with $D = 10^{40}$ and $L \geq 0$, the square-root endpoints are

$$\frac{\lfloor \sqrt{LD} \rfloor}{D}, \quad \frac{\lceil \sqrt{UD} \rceil}{D}.$$

Integer square roots and exact square comparisons give these bounds.

Logarithms. For each rational argument $z > 0$, use exact powers of two to write $z = 2^k y$, $1 \leq y < 2$. With $t = (y-1)/(y+1)$ and $N = 64$,

$$\log y = 2 \sum_{j=0}^{N-1} \frac{t^{2j+1}}{2j+1} + R, \quad 0 \leq R \leq \frac{2t^{2N+1}}{(2N+1)(1-t^2)}. \quad (9)$$

This follows from the scalar logarithm series by replacing all omitted denominators by $2N+1$ and summing a geometric series. Since $0 \leq t < 1/3$, the bound applies; the same formula at $y = 2$ encloses $\log 2$. Then $\log z = k \log 2 + \log y$ is enclosed, with the sign of k handled by interval multiplication. Scalar monotonicity encloses a logarithm on an interval by applying this procedure to its two endpoints.

Exponential series. For $j \in \{0, 1\}$, let $a_n = q^n/(2n + j)!$ and sum through $n = N = 64$. Set

$$R_j = \frac{q}{(2N + 2 + j)(2N + 1 + j)}, \quad 0 \leq \sum_{n=N+1}^{\infty} a_n \leq \frac{a_{N+1}}{1 - R_j}. \quad (10)$$

Indeed, every successive ratio after a_N is at most R_j . Both computed q intervals are positive and have upper endpoint less than 11, so $R_j < 1$ and the geometric bound is valid. Finite terms and both remainder bounds (9)–(10) are themselves evaluated by outward rational interval operations. These inequalities justify enclosure of the analytic functions, independently of a numerical transcendental library.

Table 1 records intermediate enclosures, rounded outward to twelve decimal places from the finer internal intervals. The final trace is calculated with the internal intervals; the table is a record, not a replacement of those internal endpoints.

Table 1: Exact rational enclosures for the two evaluations.

Quantity	Input interval	Output interval
$b(u)$	[3.267724418239, 3.267724418240]	[1.560431310855, 1.560431310856]
$b(v)$	[2.279046405316, 2.279046405317]	[2.359684013644, 2.359684013645]
q	[10.051320716990, 10.051320716991]	[3.573600232969, 3.573600232970]
c	[2.403602341720, 2.403602341721]	[0.032277981463, 0.032277981464]
$\sqrt{(1-u)/(1-v)}$	[0.360097666849, 0.360097666850]	[2.434169955874, 2.434169955875]
$C(q)$	[11.929281600241, 11.929281600242]	[3.386503968356, 3.386503968357]
$H(q)$	[3.749483440794, 3.749483440795]	[1.711541207712, 1.711541207713]

Substitution in (7) gives (5). In particular,

$$T(\Phi(\rho), \Phi(\sigma)) - T(\rho, \sigma) \geq 0.836799311148 > 0.835 > 0.$$

Strict monotonicity of the scalar logarithm proves the last assertion of Theorem 1, completing the counterexample.

Reproducibility and assistance. The accompanying Python standard-library verifier evaluates exactly one state pair and one channel. From the supplement’s root directory, run `python3 manuscript/reproduce.py`; it checks the rational invariants, determinants, intermediate enclosures, and strict separation without overwriting the stored certificate. Its interval engine is provided in `evidence/certify_witness.py`. Floating-point calculations assisted discovery only. A separate verification pass reconstructed the matrix invariants and checked the trace intervals by a rational positive-series calculation. AI assistance was used in discovery, proof drafting, verifier development, manuscript preparation, and that separate review. A separate model context is not external peer review or formal proof-assistant checking. The analytic reduction and remainder estimates above explain the exact computational certificate. No classification of other states, channels, dimensions, or orders is claimed.

References

- [1] Zijian Zeng, Houde Liu, and Kurunathan Ratnavelu, *A Counterexample to the Tang–Zhang Schatten Norm Conjecture and Sharp Positive Results*, Preprints (2026).
doi:10.20944/preprints202608.1067.v1.